

Supplementary Material

Title: Quantisation and seasonal-boundary artefacts in Sentinel-2 rice phenology: a reproducible quality-control framework for cyclone-impact studies

Authors: Supranab Panda, Sarat Chandra Sahu

S1. Extended Methods

S1.1 Dekadal compositing protocol

Sentinel-2 L2A scenes were collected from GEE collection COPERNICUS/S2_SR_HARMONIZED for the period 2019–2024. For each district polygon, valid-pixel NDVI values were aggregated into 10-day (dekadal) median composites. The composite boundaries follow the standard FAO dekadal partitioning: dekad 1 = DOY 1–10, dekad 2 = DOY 11–20, dekad 3 = DOY 21–end of month (28/29/30/31 depending on month). For the Kharif window (approximately DOY 150 to DOY 395 in the v2.0 extended window), this yields 25 dekadal bins per pixel per year.

A pixel was excluded from a dekadal composite if fewer than 2 valid SCL-passing observations were available within the 10-day window. District-level NDVI for each dekad was computed as the median of all valid pixel composites within the district WorldCover cropland mask. Missing district-dekad values (no valid pixels) were flagged as NaN and passed to the Whittaker smoother for gap-filling.

S1.2 Whittaker smoother parameters

The Whittaker smoother (Eilers, 2003) was applied to each pixel's dekadal NDVI series prior to double-logistic fitting. The smoothing parameter λ was selected by generalised cross-validation (GCV), with the GCV search grid spanning $\lambda \in \{1, 5, 10, 50, 100, 500, 1000, 5000\}$. For pixels with fewer than 8 valid observations in the Kharif window (out of 25 possible dekads), the GCV was suppressed and a fixed $\lambda = 100$ was applied. Pixels with fewer than 5 valid observations were excluded from phenometric extraction entirely (flagged as `fit_fail = 1`).

S1.3 Double-logistic fitting details

The double-logistic function (Beck et al., 2006) was fitted by the Levenberg–Marquardt non-linear least squares algorithm (`scipy.optimize.curve_fit`, Python). Initial parameter estimates were set as: $c_1 = \text{NDVI}_{\min}$, $c_2 = \text{NDVI}_{\max} - \text{NDVI}_{\min}$, $k_1 = k_2 = 0.1$, $t_1 = \text{DOY } 185$, $t_2 = \text{DOY } 290$. The Kharif-window SOS and EOS were extracted at the 20% of amplitude threshold; POS at the maximum of the fitted curve. Fits with a root-mean-squared residual > 0.15 NDVI units were flagged as `fit_quality_fail` (Gate C).

S1.4 v2.0 vs. v1.0.2 fitting-window change

The critical difference between the v1.0.2 and v2.0 pipelines is the extension of the EOS fitting window. In v1.0.2, the Kharif window ended at DOY 365 (31 December), creating a hard

boundary at the last valid dekad of the calendar year (DOY 351, dekad 35); the empirical EOS mode in the v1.0.2 panel falls at DOY 349 (the centre of that boundary-adjacent dekad). In v2.0, the window is extended to DOY 395 (approximately 4 February of the following year) by including the January dekads of year $t+1$ in the fit for year t . This 30-day extension is sufficient to accommodate the senescence limb for late-harvesting varieties (harvest in December) without contaminating the window with the subsequent dry-season or rabi vegetation signal (which begins in March in Odisha). The extension has no effect on SOS extraction (bounded by DOY 155 lower limit in Gate B) and minimal effect on POS extraction.

S2. Cyclone Climatology Note

The three treatment cyclones (Fani 2019, Amphan 2020, Yaas 2021) represent an unusual clustering of pre-Kharif Bay of Bengal landfalls within a three-year window. The 1990–2024 IBTrACS record for the Bay of Bengal shows that major cyclone landfalls on the Odisha coast in the May window (pre-monsoon) occur approximately once per 5–7 years on average. The 2019–2021 cluster is statistically unusual at approximately the 5% level under a Poisson model for landfall timing (exact test: $p \approx 0.04$ for three consecutive pre-Kharif events within a six-year window). This clustering is a feature of the natural experiment, not a sampling artefact: the consecutive events provide multiple treatment observations that increase the effective sample size of the DiD estimator, while the consecutive nature within the panel creates the potential for cumulative adaptation effects discussed in Section 5.4.

Figure S1 shows the spatial distribution and intensity tracks of all Bay of Bengal cyclones in the IBTrACS archive from 1990 to 2024 that made landfall within 200 km of the study area, providing the climatological context for the three treatment events.

S3. Supplementary Tables

Table S1. Panel-wide phenometric mode-shares, v1.0.2 vs. v2.0

Metric	v1.0.2 panel n	v1.0.2 mode DOY	v1.0.2 mode- share	v2.0 panel n	v2.0 unique DOYs	v2.0 mode- share	Gate A (≤ 0.20)
SOS	64	196	0.203	48	36	0.083	Pass
POS	64	288	0.656	48	33	0.083	Fail – Pass
EOS	44*	349	0.727	48	38	0.083	Fail – Pass

The v1.0.2 EOS panel has $n = 44$ (rather than 64) because 20 district-year cells were censored due to double-logistic fitting failures at the November-end window edge. Sources: `baci_panel_real_v1.csv` (pipeline = ‘raw’) and `baci_panel_real_v22.csv`. The headline finding is the collapse of EOS mode-share from 0.727 to 0.083 and POS mode-share from 0.656 to 0.083 — two distinct quantisation artefacts caught by Gate A.

Table S2. Event-study coefficients with 95% CI, v2.0 panel

Metric	k	$\hat{\beta}$ (d)	SE	CI 95% lo	CI 95% hi	n_treat
SOS	0 (ref)	0.00	—	—	—	5
SOS	1	+19.70	12.58	−4.95	+44.35	5
SOS	2	−0.57	11.54	−23.18	+22.05	5
SOS	3	+16.27	16.19	−15.47	+48.00	5
SOS	4	+1.60	14.96	−27.72	+30.92	5
SOS	5	−21.40	18.79	−58.23	+15.43	5
POS	0 (ref)	0.00	—	—	—	5
POS	1	−4.30	7.21	−18.43	+9.82	5
POS	2	−0.67	9.28	−18.86	+17.52	5
POS	3	−1.20	8.19	−17.24	+14.84	5
POS	4	+5.17	8.19	−10.89	+21.22	5
POS	5	−1.97	6.74	−15.17	+11.24	5
EOS	0 (ref)	0.00	—	—	—	5
EOS	1	+3.40	7.04	−10.40	+17.20	5
EOS	2	+8.67	8.18	−7.37	+24.71	5
EOS	3	+9.27	7.34	−5.12	+23.65	5
EOS	4	−1.70	7.06	−15.54	+12.13	5
EOS	5	+16.83	13.04	−8.72	+42.39	5

Reference: k = 0 (2019, Cyclone Fani year). Coastal (treatment) districts: BLS, BHA, KDP, JGS, PUR (n_treat = 5). SE and CI from cluster-robust standard errors (G = 8 districts). No WCB is applied to event-study coefficients.

Notes: k indexes years since 2019 (k=1 = 2020, Amphan year; k=2 = 2021, Yaas year; k=3–5 = control years 2022–2024). No pre-treatment period available; identification rests on inland-control year-FE absorption. Source: analysis/v22/results/event_study_v22.csv.

S4. Cell-Level DiD Robustness Check

To confirm that the null static DiD result is not an artefact of district-level aggregation, we re-estimated Equation (1) at cell-year resolution. The cell-level panel comprises n = 1,223 fit_ok cell-year observations across 436 unique cells nested in 8 districts, spanning Kharif seasons 2019–2024. Inference uses wild-cluster bootstrap clustered on district (G = 8, 499 Rademacher draws), with standard errors clustered on district to respect within-district correlation across cells and years.

Table S3. Static DiD ATT estimates at cell-year resolution (v2.0 QC-passing panel).

Metric	$\hat{\tau}$ (d)	SE	n_obs	n_cells	p_cluster	p_wcb
SOS	+4.91	4.96	1,223	436	0.322	0.232
POS	-2.58	1.73	1,223	436	0.135	0.096
EOS	-4.44	3.64	1,223	436	0.223	0.250

Source: *analysis/v22/results/did_cell_level_v22.csv*. WCB inference: 499 Rademacher draws, cluster on district ($G = 8$).

All three cell-level point estimates are directionally consistent with the district-level analysis (Table 1 of the main text) and similar in magnitude (within 3 days for all three metrics). None of the cell-level estimates cross the 5% significance threshold despite the $\sim 25\times$ larger effective sample. The closest case is POS, with $p_{wcb} = 0.096$, which is a soft indication of within-district heterogeneity but does not meet conventional thresholds for statistical significance. This robustness check supports the interpretation in the main text: the QC framework removes the boundary-artefact variance that drove the apparent v1.0.2 EOS effect, and there is no detectable cyclone signal at either aggregation level in this 8-district, 3-cyclone-event design.

The full cell-level event-study output is archived at *analysis/v22/results/event_study_cell_v22.csv* in the v2.0-refit branch of the code repository.

S5. Parallel-trends event study, selection bounds, and power analysis

This section reports the rebuttal-evidence tables generated from *rse_final/reviewer_rebuttal_analysis.py* and supports §3.5, §4.1, and §4.6 of the main text.

S5.1 Table S10 — Event-study leads/lags with 2018 as reference year

The event-study specification of Equation (2) is re-estimated with 2018 as the reference year, so that 2017 enters as a single pre-period lead and 2019–2024 enter as post-period lags. Coefficients (in days), 95% confidence intervals from cluster-robust standard errors, and p-values are reported by metric. EOS coefficients cannot be estimated because the v2.0 panel EOS distribution takes only two distinct values (DOY 349/350).

Metric	Year	Period	β (d)	SE	95% CI low	95% CI high	p
SOS	2017	pre	+63.60	47.43	-29.36	+156.56	0.180
SOS	2019	post	+76.00	56.00	-33.75	+185.75	0.175
SOS	2020	post	+96.27	61.18	-23.65	+216.18	0.116
SOS	2021	post	+57.20	49.68	-40.16	+154.56	0.250
SOS	2022	post	+133.20	45.17	+44.66	+221.74	0.003
SOS	2023	post	+76.00	51.16	-24.27	+176.27	0.137
SOS	2024	post	+33.20	47.33	-59.57	+125.97	0.483
POS	2017	pre	+2.40	13.91	-24.85	+29.65	0.863

Metric	Year	Period	β (d)	SE	95% CI low	95% CI high	p
POS	2019	post	+2.40	13.91	-24.85	+29.65	0.863
POS	2020	post	-10.00	10.69	-30.95	+10.95	0.350
POS	2021	post	-10.00	10.69	-30.95	+10.95	0.350
POS	2022	post	-10.00	10.69	-30.95	+10.95	0.350
POS	2023	post	0.00	18.52	-36.29	+36.29	1.000
POS	2024	post	-3.80	12.92	-29.13	+21.53	0.769

The Wald test of joint zero on the pre-period lead (1 d.f.) is $\chi^2 = 1.80$, $p = 0.18$ for SOS and $\chi^2 = 0.03$, $p = 0.86$ for POS. Neither rejects the parallel-trends null at conventional levels, although the test is severely underpowered with a single pre-period observation.

S5.2 Table S10b — Landsat-5/7/8 pre-period extension (camera-ready)

The Landsat-5/7/8 harmonised extension module (`gee/Module_13_Landsat_Pretrends.js`) produces a four-year pre-period extension (2015–2018) of the SOS, POS, and EOS panel. Roy et al. (2016) Landsat–Sentinel-2 NDVI cross-calibration coefficients are applied. The corresponding placebo-DiD module is `rse_final/module_14_pretrend_placebo.py`. Module 13 is integrated into the released codebase; the four-year pre-trends Wald test will be appended as Table S10b in the camera-ready version.

S5.3 Table S11 — MAR vs. MNAR diagnostic

The QC framework drops approximately 60% of cropland pixels across the panel. We test whether those drops are systematically more frequent in cyclone-exposed coastal districts than in inland controls by estimating Equation (1) with `fit_fail_rate` (the per-district-year share of cropland pixels that fail any QC gate) as the outcome on the full unfiltered 48-row panel.

Outcome	DiD coefficient	SE	95% CI low	95% CI high	p	Verdict
<code>fit_fail_rate</code>	-0.0089	0.0233	-0.0546	+0.0367	0.702	MAR (no selection bias)

The $\text{treat} \times \text{post}$ effect on QC fit-failure rate is statistically indistinguishable from zero and the 95% CI rules out any treatment-induced increase in fit-failure rate larger than ~ 4 percentage points. We interpret this as direct evidence that the QC procedure is MAR with respect to treatment.

S5.4 Table S12 — Manski and Lee selection bounds

As a non-parametric worst-case check, we compute Manski (1990) worst-case bounds and Lee (2009) trimming bounds on each ATT. The Manski bounds use the biologically plausible phenometric windows (Gate B) as the worst-case range for QC-dropped cells. The Lee bounds use the observed trimmed quantile of the QC-passing cells.

Metric	$\hat{\tau}$ main	SE	Manski lo	Manski hi	Lee lo	Lee hi	Robust null (Manski)	Robust null (Lee)
SOS	+7.56	8.10	+0.29	+14.83	-1.78	+16.90	No (just)	Yes
POS	-2.32	2.91	-7.47	+2.83	-8.94	+4.30	Yes	Yes
EOS	-4.11	4.62	-12.72	+4.50	-15.16	+6.94	Yes	Yes

The POS and EOS nulls are robust under both Manski and Lee bounding assumptions. The SOS null is robust under Lee trimming; the Manski lower bound just excludes zero on the lower side, reflecting the wider point estimate for SOS rather than a violated MAR assumption.

S5.5 Table S13 — Balanced-pixel sub-panel (camera-ready)

A balanced-pixel sub-panel restricted to pixels that pass QC in every year from 2019 through 2024 removes pixel entry/exit from the panel as a possible source of imbalance. The balanced-pixel export is delivered through Module 04b of the released GEE codebase; the camera-ready version will report the balanced-pixel $\hat{\tau}$ coefficients alongside Model 1 and Model 2 of the main text.

Metric	Balanced-pixel $\hat{\tau}$	SE	Δ from main $\hat{\tau}$	Note
SOS	(pending)	(pending)	(pending)	Run Module 04b balanced-pixel export and rerun module_14_pre_trend_placebo.py.
POS	(pending)	(pending)	(pending)	(as above)
EOS	(pending)	(pending)	(pending)	(as above)

S5.6 Table S14 — Minimum detectable effect (MDE) at 80% power

The panel-level (Model 1) and pixel-level (Model 2) MDEs at 80% power and $\alpha = 0.05$ (two-sided) are derived from the post-QC residual standard errors of the respective specifications.

Metric	$\hat{\tau}$ main	SE (district)	MDE district 80% (d)	SE (pixel approx)	MDE pixel 80% (d)
SOS	+7.56	8.10	22.68	0.209	0.59
POS	-2.32	2.91	8.15	0.075	0.21
EOS	-4.11	4.62	12.94	0.119	0.33

The district-level model (Model 1) can detect effects of ≥ 22.7 d for SOS, ≥ 8.1 d for POS, and ≥ 12.9 d for EOS at 80% power. The pixel-level model (Model 2) has an MDE below 1 d for all three

metrics. The pixel-level null therefore actively rules out cyclone effects on the QC-passing pixel universe larger than ± 1 d.

S6. Open Science Framework Pre-registration

The study was pre-registered on the Open Science Framework prior to any data analysis. The pre-registration documents: - The primary research question (phenometric boundary artefact characterisation) - The QC gate definitions and thresholds - The DiD estimating equation (Equation 1 of the main text) - The three phenometric outcomes (SOS, POS, EOS) - The treatment group (5 coastal districts) and control group (3 inland districts) - The treatment period (2019–2021: Fani, Amphan, Yaas) - The inference protocol (cluster-robust SE + WCB, 499 draws)

Pre-registration DOI: 10.17605/OSF.IO/C4MP8 **Pre-registration URL:** <https://osf.io/c4mp8>

One amendment was registered (2026-05-01): clarification that pre-treatment parallel-trend testing is unavailable due to the panel beginning in the first treatment year (2019), and documentation of the inland-control identification strategy as the primary basis for the DiD identifying assumption.